Function Context: B2

Function Background

This function comes from Requests' HTTP request-preparation logic. An HTTP request body may come from JSON, ordinary form data, files, or streamed data. The function constructs the body according to the input form and sets related request headers when necessary.

Input and Output

Inputs: data, ordinary request data or iterable data; files, files to upload; json, data to be encoded as JSON.

Output: The function does not directly return a result.

Side effects: Writes the final request body to self.body; may update self.headers; and records the request-body position in some cases.

Key Terms

Term	Meaning
request body	The part of an HTTP request that carries data.
header	Metadata in an HTTP request.
JSON	A commonly used structured data format.
form data	Data submitted as key-value pairs.
streamed body	A data stream read and sent incrementally.
multipart encoding	An encoding format commonly used for file uploads.
Content-Length	The length of the request body.
Transfer-Encoding: chunked	A method for transmitting a request body in chunks.
Content-Type	The media type of the request body.

Helper Variables and Functions

Name	Brief Description
complexjson.dumps(...)	Encodes an object as a JSON string.
super_len(data)	Attempts to determine the data length.
_encode_files(...)	Encodes files and ordinary data into a request body.
_encode_params(...)	Encodes ordinary parameters as form data.
prepare_content_length(body)	Sets length-related information based on the request body.
tell()	Gets the current read position of streamed data.

Background Not Required

Participants do not need to master all details of the HTTP protocol. They only need to determine how different input forms affect the request body and request headers based on the function's logic.